

Week 9 - Groups, rings, and Cayley's theorem

On the first mock exam, you encountered the following mathematical object.

Definition: Suppose S is a set, and for all $a, b \in S$ there is a specifically chosen element $a \square b \in S$ such that for all $a, b, c \in S$, we have the identity $(a \square b) \square c = a \square (b \square c)$. Let's call such a pair $(S, \square)$ satisfying these properties a **soup**.

1) Is every soup an example of a group? If so, explain why. If not, identify which property(s) of groups that a soup might not satisfy, and give an example of a soup that is not a group.

2) Is every group an example of a soup? If so, explain why. If not, identify which property(s) of soups that a group might not satisfy, and give an example of a group that is not a soup.

2) (a) Find an example of a $(G, \mu, e, \square)$ such that (G, μ, e) is a group, $(G, \square)$ is a soup, and the group and soup structures satisfy the following relationship: for all $a, b \in G$, we have

$$a \square (\mu(b, c)) = \mu(a \square b, a \square c).$$

Hint: experiment with changing the notations of $\mu, \square, e$ with more familiar things, and then identify which should be which.

(b) Does your example in (a) satisfy the following property: there exists an element $I \in G$ such that $I \square g = g = g \square I$ for all $g \in G$? If so, give an example that does not. If not, give an example that does.

(c) Consider your example that *does* satisfy the property in (b). Does this satisfy the following additional property: for all $g \in G$ with $g \neq e$, there exists $h \in G$ such that $hg = I = gh$? If so, give an example that does not. If not, give an example that does.

(d) Consider your example that satisfies both of the properties in (b) and (c). Is $(G, \square, I)$ a group? Is $(G - \{e\}, \square, I)$ a group? Prove why or why not for both.

3) Let $S \subset \mathbb{R}^2$ be some subset of the plane (think of shapes like: square, circle, octagon, or draw your own!). Let G be the set of bijections $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that for all $s \in S$ we have $f(s) \in S$ and $f^{-1}(s) \in S$. Prove that G is a group with composition of functions as the operation μ . Such a group G can be thought of as the *group of symmetries* of S .

The rest of this worksheet will be dedicated to proving **Cayley's Theorem**, which states that “every group is a group of symmetries”. In this sense, group theory is the study of symmetry.

4) Let (G, μ, e) be any group. For an element $a \in G$ let $\ell_a : G \rightarrow G$ be the function defined by $\ell_a(g) = \mu(a, g)$. Prove that ℓ_a is a bijection.

5) If A is a set, let S_A be the **symmetric group** of A , that is, the group of bijections $A \rightarrow A$ under composition. Prove that the function $L : G \rightarrow S_G$ defined by $L(a) = \ell_a$ is injective.

6) Let (G, μ, e) and (G', μ', e') be groups. Recall that a **group homomorphism** from G to G' is a function $\varphi : G \rightarrow G'$ such that

$$\varphi(\mu(g, h)) = \mu'(\varphi(g), \varphi(h))$$

for all $g, h \in G$. Show that the function L from problem (5) is a group homomorphism.

7) A group homomorphism $\varphi : G \rightarrow G'$ is called an **isomorphism** if it is bijective and φ^{-1} is also a group homomorphism. We say that G and G' are **isomorphic** if there exists an isomorphism between them. Prove that a bijective group homomorphism is always an isomorphism.

8) Prove **Cayley's theorem**: Every group is isomorphic to a subgroup of the symmetric group of some set.